

**Comparative Analysis of Gene Expression in Surgically Removed Lung Adenocarcinoma
and Normal Adjacent Lung Tissues.**

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Introduction:

Lung adenocarcinoma is a type of non-small cell lung cancer (NSCLC) which occurs on the outer cellular lining of the lung. NSCLC tumors are typically found in the alveoli, where gas exchange occurs. Lung adenocarcinoma accounts for nearly 40% of all NSCLCs and is the most common lung cancer for non-smokers (“Adenocarcinoma of the Lung”). Improving our understanding of the genetic mutations and mechanisms in lung adenocarcinoma is essential for advancing diagnostics and therapeutics to treat NSCLCs.

Utilizing RNA sequencing (RNA-seq) is key to understanding significant transcriptomic changes between tumor and normal tissues. Previous differential gene expression and pathway-based analyses that used RNA-seq data have identified key biological processes involved in lung adenocarcinoma, including epithelial-to-mesenchymal transition (EMT), extracellular matrix remodeling, and dysregulation of cell cycle pathways (Kalluri & Weinberg, 2009). These approaches provide insight into both tumor biology and potential molecular targets for therapeutic development.

Environmental exposures, particularly tobacco smoking, also heavily influence the lung cancer transcriptome. It has been previously reported that tobacco-related NSCLC takes on a 10-fold increased mutational frequency between smokers and non-smokers (Govindan et al., 2012). In this study, RNA-seq data from the GSE40419 dataset was analyzed to characterize differential gene expression between lung adenocarcinoma tissue and matched normal lung tissue. Additionally, the impact of smoking status on tumor-associated gene expression was evaluated through interaction modeling and subgroup analyses to better understand its role in lung adenocarcinoma mechanisms.

Materials and Methods:

Using the GSE40419 dataset, RPKM RNA-seq data for 87 surgical lung adenocarcinomas and 77 surgical normal lung tissue samples from a Korean population were analyzed for differential gene expression. Table 1 displays the sample size of smoker status individuals by tissue of all 164 samples. The 87 lung adenocarcinoma samples “were analyzed by transcriptome sequencing combined with whole-exome (n = 76) and transcriptome (n = 77) sequencing of matched normal lung tissue samples” (Seo et al., 2012).

In this study, the RNA-seq data was cleaned and filtered for downstream analysis. Specifically, duplicate gene symbols were aggregated by taking the mean expression across entries. The genes were then filtered to retain genes with expression values greater than 1 in at least 20% of samples in order to remove lowly expressed genes. Then, the data was log2-transformed with a pseudo-count of 1.

Following the filtration and transformation, principal component analysis (PCA) was performed on the scaled log2-transformed expression values of tissue groups by smoking status. Differential gene expression was then performed on paired samples, including an analysis on the interaction between smoking status and tissue type. Only patients with both tumor and matched normal samples were retained ($n = 77$ pairs). Following the analysis, Gene Set Enrichment Analysis (GSEA) was performed with genes ranked by interaction log2 fold change to identify biological pathways associated with tumor-specific transcriptional changes. Gene symbols were mapped to Entrez IDs using org.Hs.eg.db, and Gene Ontology (GO) enrichment analysis was performed using a hypergeometric test with all tested genes as the background universe.

Differential gene expression analysis was performed using the *limma* package in R. Linear models were fitted to the log2-transformed expression data, and statistical significance was assessed using empirical Bayes moderated t-tests to stabilize the variance estimation. For paired analyses, patient identity was included as a blocking factor in the design matrix ($\sim$ Patient + Tissue). For interaction analysis, the linear model included a tissue-by-smoking interaction term by patient identity ($\sim$ Patient + Tissue $\times$ Smoking_Group) to evaluate whether smoking status impacts tumor-associated gene expression changes. For subgroup analyses, contrast-based linear models were used to estimate the tissue type differences within each smoking status group. P-values were adjusted for multiple testing using the Benjamini–Hochberg (BH) false discovery rate (FDR) method.

Following all differential gene analyses, the top three upregulated and downregulated differentially expressed genes (DEGs) were also generated for each comparison and incorporated into Table 2. P-value histograms and QQ plots were used to assess statistical calibration for all models as well.

OpenAI's ChatGPT model GPT-5.3-mini was used to guide the creation of R code to generate figures, perform PCA and GSEA analysis, and use of *limma* for differential gene

expression analysis. The code for this project can be found at this link:

https://github.com/bluker17/Lung_Adenocarcinoma_Analysis

Results:

Global Transcriptomic Profile

Principal component analysis (PCA) revealed clear separation between lung adenocarcinoma and normal lung tissue samples (Figure 1). The first two principal components accounted for 30.4% of the total variance (PC1 = 20.1%, PC2 = 10.3%). Tumor samples clustered distinctly from normal samples, with partial overlap observed across smoking and non-smoking subgroups.

Paired Patient Sample Differential Gene Expression Analysis

In the primary patient tissue sample paired analysis ($n = 77$ matched pairs), 1,265 genes were significantly differentially expressed between tumor and normal lung tissue ($\text{FDR} < 0.05$, $|\log_2\text{FC}| > 1$). Of these, 467 genes were upregulated and 798 genes were downregulated in tumor tissue (Figure 2). The three most differentially expressed upregulated genes were *SPP1*, *PYCR1*, and *ALDH18A1*. The three most differentially expressed downregulated genes were *GPM6A*, *CCBE1*, and *RADIL* (Table 2).

Subsequent gene ontology enrichment analysis of significant DEGs identified multiple significantly enriched biological processes related to angiogenesis regulation, cellular migration, and extracellular matrix organization (Figure 3).

Smoking-Dependent Interaction Effects

Interaction analysis of smoking status and paired tissue samples identified 28 genes with significant smoking-dependent effects on tumor-associated gene expression ($\text{FDR} < 0.05$). 9 genes showed positive smoking-dependent interaction effects, while 19 showed negative interaction effects. (Figure 4). The most significant upregulated genes included *CDC20*, *PRAME*, and *MYBL2*, while the most significant downregulated genes included *SCTR*, *C1orf116*, and *ABCC3* (Table 2).

According to Figure 5, GSEA revealed genes in biological processes, such as chromatid and chromosome segregation, and nuclear division, being positively enriched in tumor tissue compared to normal tissue.

Exploratory Differential Gene Analysis

In all patient samples ($n_{\text{tumor}} = 87$, $n_{\text{normal}} = 77$), 1,300 DEGs between tumor and normal lung tissue were identified ($\text{FDR} < 0.05$, $|\log_2\text{FC}| > 1$). Of these, 487 genes were upregulated and 813 genes were downregulated in tumor tissue (Figure 6).

In non-smokers ($n_{\text{tumor_non-smoker}} = 36$, $n_{\text{normal_non-smoker}} = 34$), 1,136 genes were differentially expressed between non-smoker tumor and normal lung tissue ($\text{FDR} < 0.05$, $|\log_2\text{FC}| > 1$). Of these, 438 genes were upregulated and 698 genes were downregulated in tumor tissue (Figure 7).

In smokers ($n_{\text{tumor_smoker}} = 47$, $n_{\text{normal_smoker}} = 40$), 1,559 genes were differentially expressed between smoker tumor and normal lung tissue ($\text{FDR} < 0.05$, $|\log_2\text{FC}| > 1$). Of these, 563 genes were upregulated and 996 genes were downregulated in tumor tissue (Figure 8).

The smoker subgroup exhibited a greater number of DEGs compared to the non-smoker subgroup. This may reflect an increased gene mutational frequency in tumors associated with smokers compared to non-smokers.

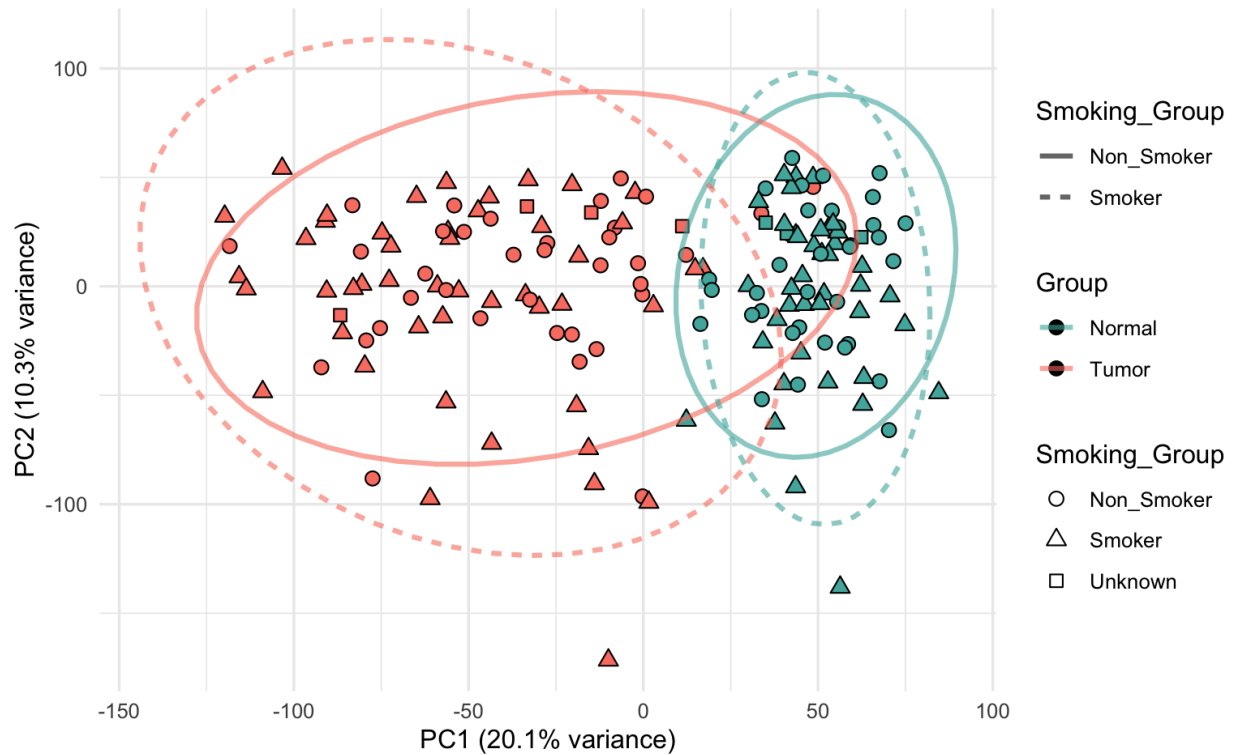


Figure 1. PCA reveals distinct transcriptome patterns of lung adenocarcinoma tissue and normal adjacent lung tissue. PCA plot of RPKM RNA-seq data from lung adenocarcinoma tissue and normal adjacent lung tissue ($n_{\text{tumor_smoker}}=47$, $n_{\text{tumor_non-smoker}}=36$, $n_{\text{tumor_unknown}}=4$, $n_{\text{normal_smoker}}=40$, $n_{\text{normal_non-smoker}}=34$, $n_{\text{normal_unknown}}=3$). Each point represents an individual sample, teal points indicate the normal sample population, and red points indicate the tumor sample population. Triangles indicate the smoker sample population, circles the non-smoker sample population, and squares the unknown sample population. PC1 accounts for 20.1% of the variance between groups. PC2 accounts for 10.3% of the variance within groups. Dashed ellipses indicate the approximate distribution and spread of each group.

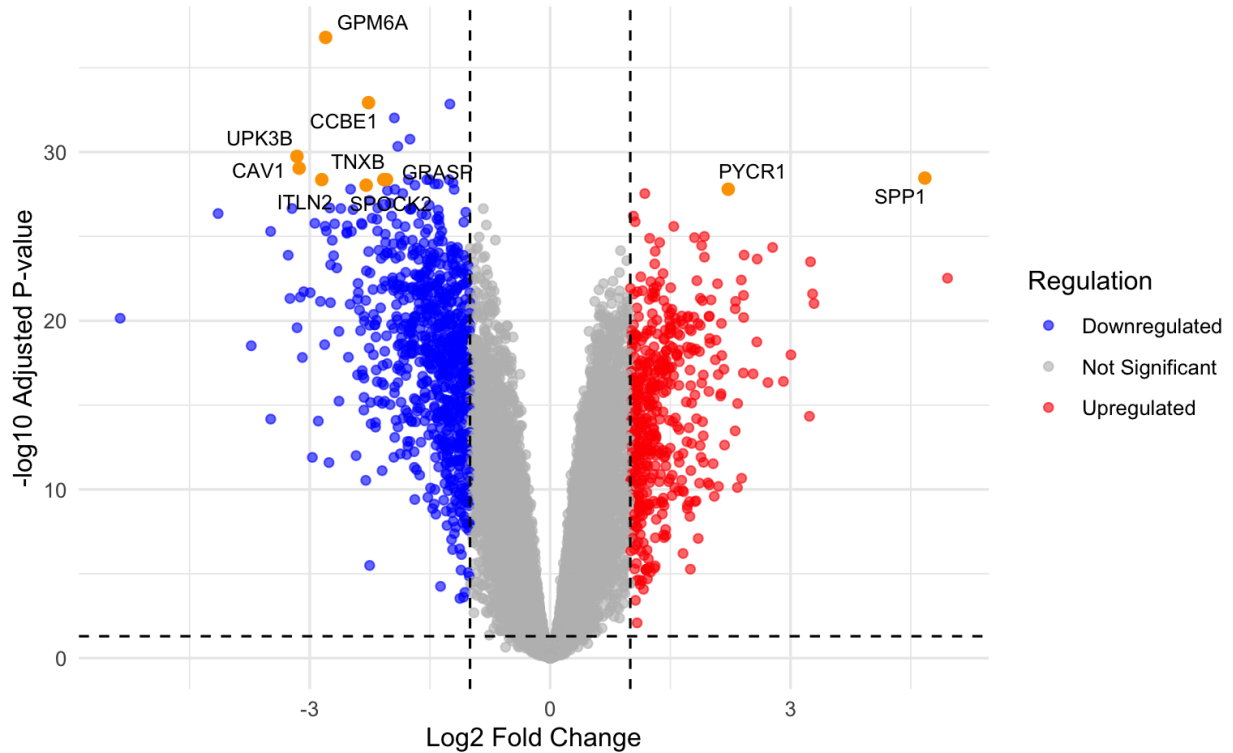


Figure 2. Volcano plot reveals DEGs of paired patient lung adenocarcinoma tissue and normal adjacent lung tissue samples. Volcano plot of log transformed RNA-seq data from paired lung adenocarcinoma tissue and normal adjacent lung tissue ($n_{\text{pairs}}=77$). Paired differential expression was performed using *limma* with a linear model ($\sim \text{patient} + \text{group}$) and empirical Bayes moderated t-tests, with BH FDR correction. Each point represents an individual gene. Red dots represent upregulated genes, blue dots are downregulated genes, and gray dots are no significant difference. Orange dots represent the ten most significant DEGs. Dashed lines mark the cutoffs for significant expression ($|\log_2\text{FC}| > 1$ and adjusted p-value < 0.05).

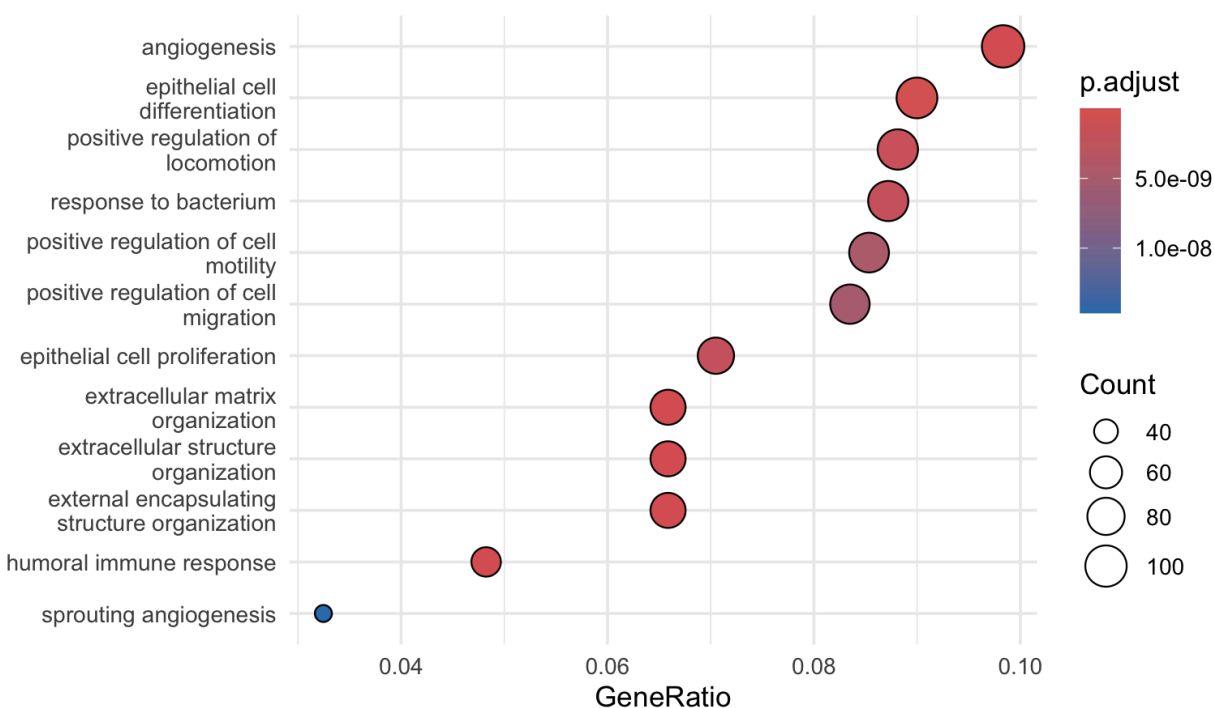


Figure 3. Gene ontology enrichment analysis of biological processes associated with paired tumor and normal tissue DEGs. GO enrichment analysis was performed on significantly differentially expressed genes identified from the paired lung adenocarcinoma vs normal adjacent lung tissue comparison. Paired differential expression was performed using *limma* with a linear model ($\sim$ patient + group) and empirical Bayes moderated t-tests, with BH FDR correction. Enrichment of Gene Ontology Biological Process (BP) terms was tested using a hypergeometric overrepresentation test, with all genes tested in the paired model used as the background universe. Each term represents a GO Biological Process category. The top 12 enriched terms are displayed. The color scale represents adjusted p-value significance (BH-FDR), with more significant terms shown in warmer colors.

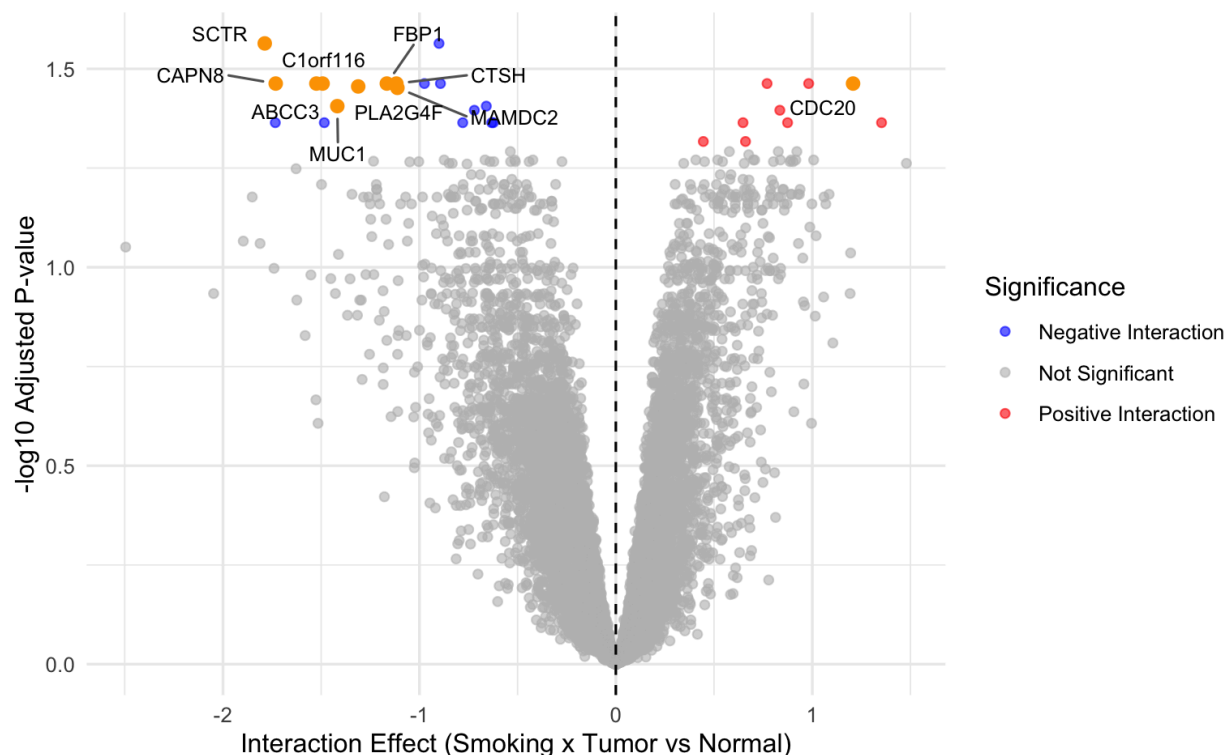


Figure 4. Volcano plot reveals genes with smoking-dependent tumor-associated expression changes in paired lung adenocarcinoma and normal adjacent lung tissue samples. Volcano plot of log-transformed RNA-seq data from paired lung adenocarcinoma tissue and normal adjacent lung tissue ($n_{\text{pairs}}=77$). Interaction analysis was performed using *limma* with a linear model ($\sim \text{Patient} + \text{Tissue} \times \text{Smoking_Group}$) and empirical Bayes moderated t-tests, with BH FDR correction. Each dot represents an individual gene. Red dots represent genes with positive interaction effects (greater tumor-associated expression changes in smokers), blue dots represent genes with negative interaction effects (reduced or reversed tumor-associated expression changes in smokers), and gray dots represent non-significant genes. Orange dots represent the ten most significant interaction genes. Dashed line marks the cutoff for no interaction effect ($|\log_2\text{FC}| > 0$ and adjusted p-value < 0.05).

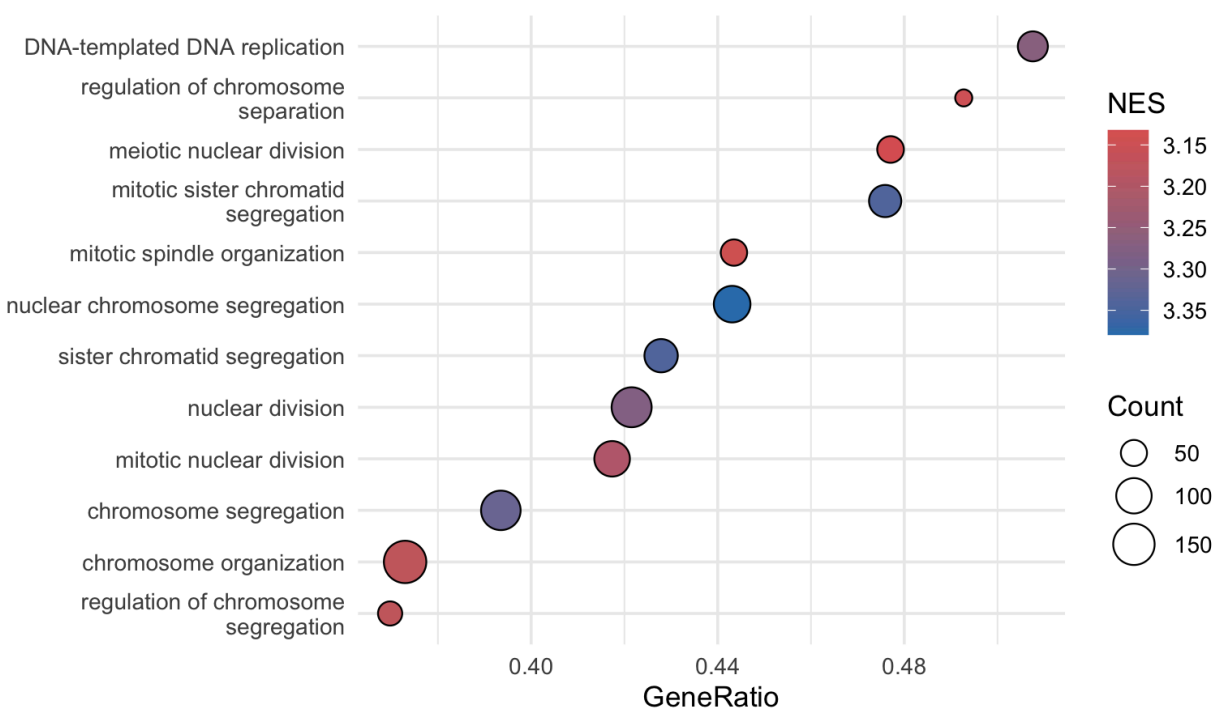


Figure 5. Gene set enrichment analysis of biological processes associated with the smoking–tissue interaction. Gene set enrichment analysis (GSEA) was performed on a ranked list of genes ordered by the interaction log2 fold change derived from a paired differential expression analysis of lung adenocarcinoma and matched normal adjacent tissue. Differential expression was modeled using *limma* with a linear model ($\sim \text{patient} + \text{group} \times \text{smoking status}$) and empirical Bayes moderated t-tests, with p-values adjusted using the BH FDR. Enrichment of Gene Ontology Biological Process (BP) terms was assessed using a permutation-based GSEA approach implemented in *clusterProfiler*. The top 12 enriched terms are shown. Color represents the normalized enrichment score (NES), with cooler color indicating more positive enrichment.

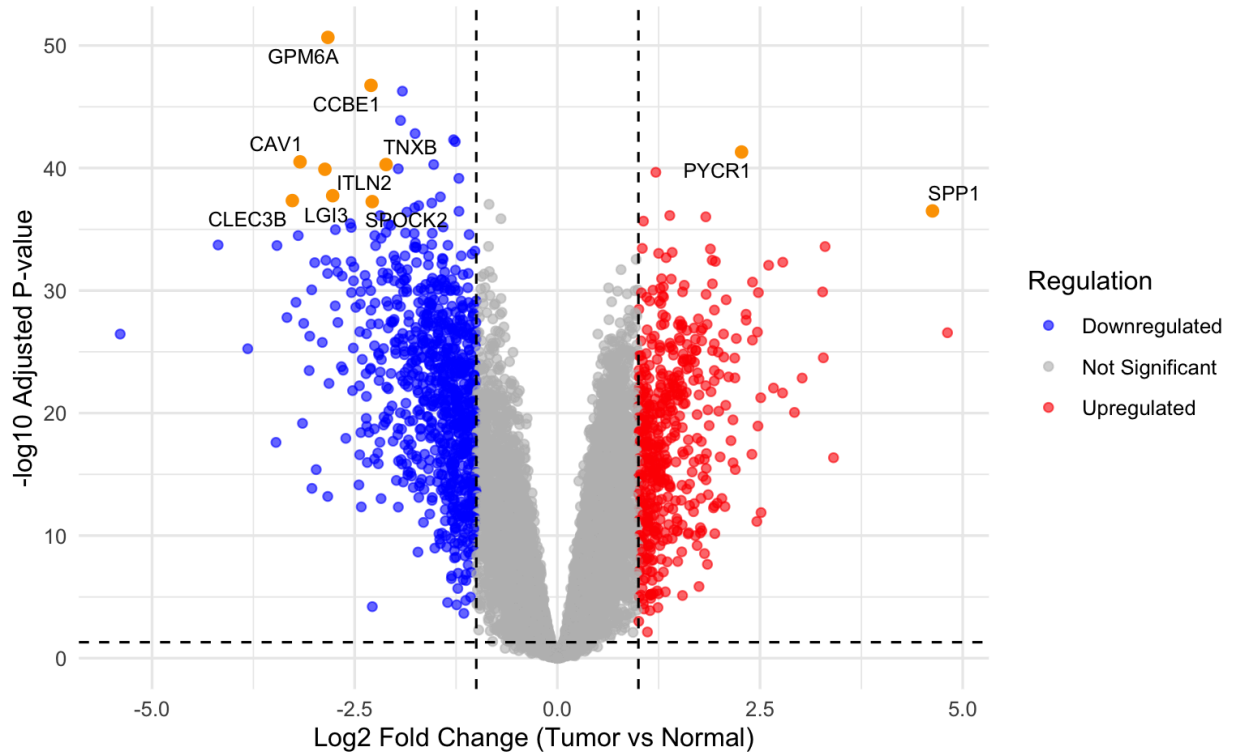


Figure 6. Volcano plot reveals DEGs of lung adenocarcinoma tissue and normal adjacent lung tissue. Volcano plot of log transformed RNA-seq data from lung adenocarcinoma tissue and normal adjacent lung tissue ($n_{\text{tumor}}=87$, $n_{\text{normal}}=77$). Differential expression analysis was performed using *limma* with a linear model ($\sim$ group) and empirical Bayes moderated t-tests, with BH FDR correction. Each dot represents an individual gene. Red dots represent upregulated genes, blue dots are downregulated genes, and gray dots are no significant difference. Orange dots represent the ten most significant DEGs. Dashed lines mark the cutoffs for significant expression ($|\log_2\text{FC}| > 1$ and adjusted p-value < 0.05).

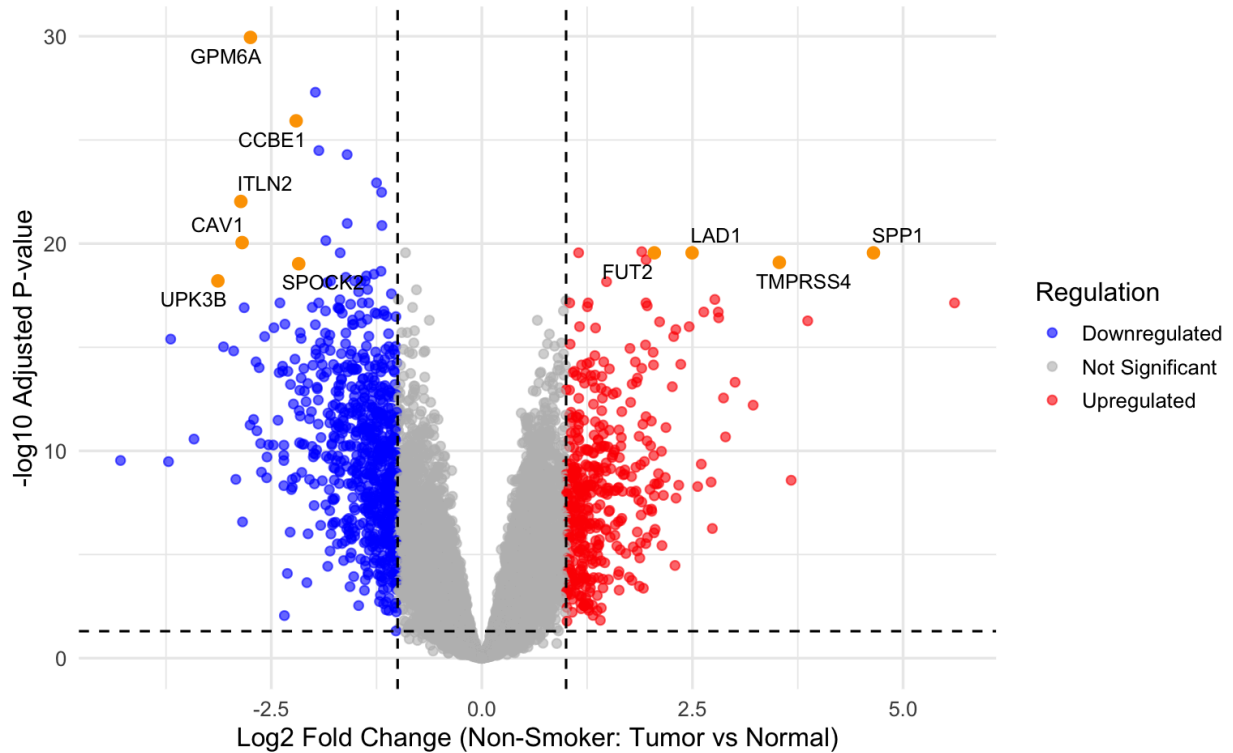


Figure 7. Volcano plot reveals DEGs of non-smoker lung adenocarcinoma tissue and non-smoker normal adjacent lung tissue. Volcano plot of log transformed RNA-seq data from non-smoker lung adenocarcinoma tissue and non-smoker normal adjacent lung tissue ($n_{\text{tumor_non-smoker}}=36$, $n_{\text{normal_non-smoker}}=34$). Differential expression analysis was performed using *limma* with a contrast-based linear model ($\sim 0 + \text{Group_Non_Smoking}$) and empirical Bayes moderated t-tests applied to the Tumor vs Normal contrast within non-smoker samples, with BH FDR correction. Each point represents an individual gene. Red dots represent upregulated genes, blue dots are downregulated genes, and gray dots are no significant difference. Orange dots represent the ten most significant DEGs. Dashed lines mark the cutoffs for significant expression ($|\log_2 \text{FC}| > 1$ and adjusted p-value < 0.05).

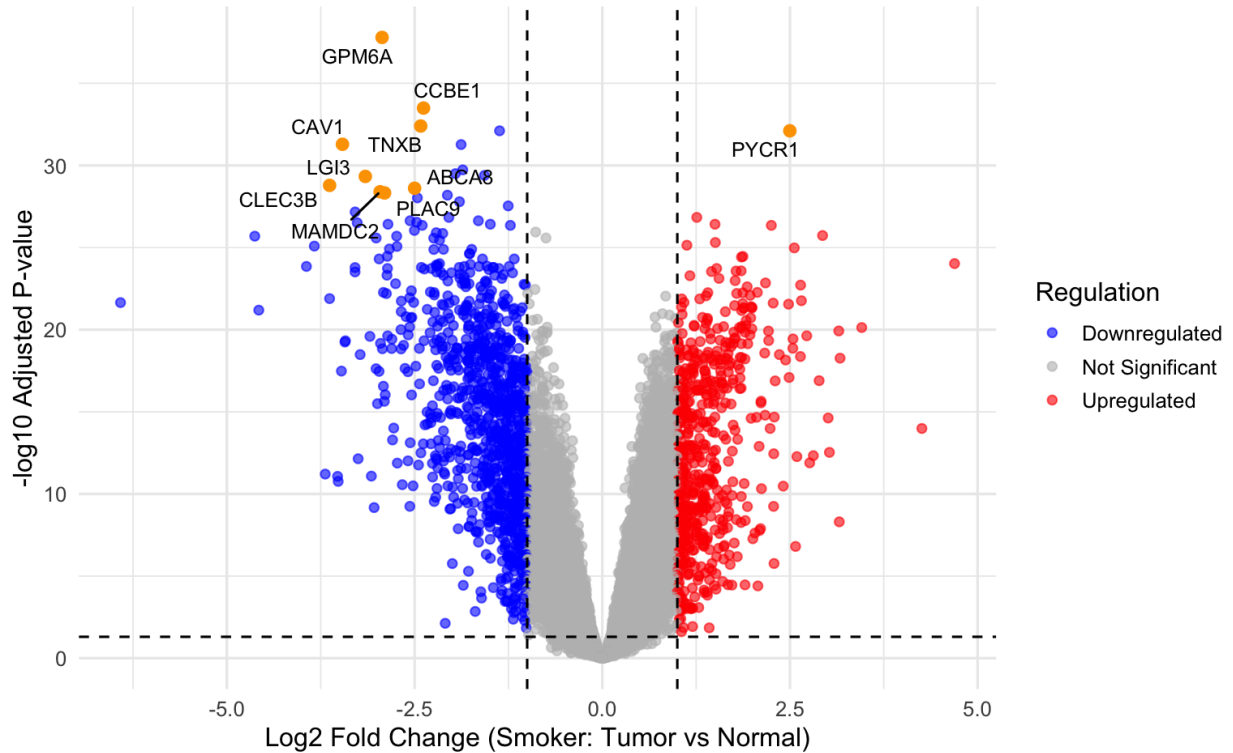


Figure 8. Volcano plot reveals DEGs of smoker lung adenocarcinoma tissue and smoker normal adjacent lung tissue. Volcano plot of RNA-seq data from smoker lung adenocarcinoma tissue and smoker normal adjacent lung tissue ($n_{\text{tumor_smoker}}=47$, $n_{\text{normal_smoker}}=40$). Differential expression analysis was performed using *limma* with a contrast-based linear model ($\sim 0 + \text{Group_Smoking}$) and empirical Bayes moderated t-tests applied to the Tumor vs Normal contrast within smoker samples, with BH FDR correction. Each point represents an individual gene. Red dots represent upregulated genes, blue dots are downregulated genes, and gray dots are no significant difference. Orange dots represent the ten most significant DEGs. Dashed lines mark the cutoffs for significant expression ($|\log_2\text{FC}| > 1$ and adjusted p-value < 0.05).

Table 1: Sample Sizes by Group and Smoking Status

Group	Non_Smoker	Smoker	Unknown
Normal	34	40	3
Tumor	36	47	4

Table 2: Top upregulated and downregulated genes identified in tumor vs normal analyses across all samples, paired samples, smokers, non-smokers, and smoking interaction analysis (* indicates genes significant in multiple comparisons)

Comparison	Regulation	gene	logFC
Tumor vs Normal (All)	Upregulated	PYCR1*	2.272296
Tumor vs Normal (All)	Upregulated	ALDH18A1*	1.214749
Tumor vs Normal (All)	Upregulated	SPP1*	4.627416
Tumor vs Normal (All)	Downregulated	GPM6A*	-2.830690
Tumor vs Normal (All)	Downregulated	CCBE1*	-2.298876
Tumor vs Normal (All)	Downregulated	C13orf36*	-1.911831
Tumor vs Normal (Non-smokers)	Upregulated	GOLM1	1.897995
Tumor vs Normal (Non-smokers)	Upregulated	ALDH18A1*	1.148396
Tumor vs Normal (Non-smokers)	Upregulated	FUT2	2.047086
Tumor vs Normal (Non-smokers)	Downregulated	GPM6A*	-2.746787
Tumor vs Normal (Non-smokers)	Downregulated	C13orf36*	-1.976093
Tumor vs Normal (Non-smokers)	Downregulated	CCBE1*	-2.204655
Tumor vs Normal (Smokers)	Upregulated	PYCR1*	2.498757
Tumor vs Normal (Smokers)	Upregulated	ALDH18A1*	1.259134
Tumor vs Normal (Smokers)	Upregulated	RCC1	1.501677
Tumor vs Normal (Smokers)	Downregulated	GPM6A*	-2.933016
Tumor vs Normal (Smokers)	Downregulated	CCBE1*	-2.380980
Tumor vs Normal (Smokers)	Downregulated	TNXB	-2.418220
Paired	Upregulated	SPP1*	4.675954
Paired	Upregulated	PYCR1*	2.222348
Paired	Upregulated	ALDH18A1*	1.181269
Paired	Downregulated	GPM6A*	-2.801642
Paired	Downregulated	CCBE1*	-2.266637
Paired	Downregulated	RADIL	-1.251265
Paired Smoker Interaction	Upregulated	CDC20	1.207580
Paired Smoker Interaction	Upregulated	MYBL2	1.352700
Paired Smoker Interaction	Downregulated	SCTR	-1.787325
Paired Smoker Interaction	Downregulated	C1orf116	-1.492882
Paired Smoker Interaction	Downregulated	ABCC3	-1.524474

Discussion:

Across multiple transcriptomic analyses, this study identified widespread gene expression differences between lung adenocarcinoma and normal adjacent lung tissue using RPKM RNA-seq data from the GSE40419 dataset.

Principal component analysis demonstrated clear separation between tumor and normal samples, indicating distinct global transcriptional profiles associated with lung adenocarcinoma tissue compared to normal lung tissue.

Differential gene expression analysis in paired samples identified 1,265 DEGs, with a greater proportion of genes downregulated in tumor tissue. Gene ontology enrichment identified biological processes related to angiogenesis regulation, cell migration, and extracellular matrix organization. These pathways were reflected in highly upregulated genes such as *SPPI* and *PYCR1*.

SPPI has been widely reported in multiple cancers, including lung cancer, and has been associated with increased cell proliferation and angiogenesis through promotion of EMT (Tang et al., 2021). EMT is a biological process in which epithelial cells acquire mesenchymal-like properties, including increased migratory capacity, extracellular matrix production, and resistance to apoptosis (Kalluri and Weinberg, 2009). *PYCR1* has also been implicated in tumor progression and may contribute to increased cell proliferation and survival through activation of the JAK/STAT signaling pathway, which regulates cellular differentiation, proliferation, and apoptosis (Shin et al., 2025).

Another key finding of this study was the identification of smoking-dependent interaction effects on tumor-associated gene expression. Although the number of interaction-associated genes was relatively small compared to overall differential expression, these genes may represent smoking-sensitive regulatory pathways that influence tumor biology. For example, *CDC20*, one of the most significant upregulated genes identified in the interaction analysis, has been implicated in cell cycle regulation and may contribute to tumor progression through effects on proliferation and apoptotic regulation in lung cancer (Xian et al., 2023). This finding is consistent with the biological processes identified in the GO enrichment analysis and GSEA, particularly those processes related to cell proliferation.

Smoking subgroup analyses further demonstrated that smokers exhibited a higher number of differentially expressed genes compared to non-smokers. This suggests that smoking may

contribute to increased transcriptional disruption or enhanced molecular divergence between tumor and normal lung tissue..

A few limitations of this study should be considered. First, the dataset is derived from a single population, which may limit generalizability to other ethnic groups. Second, smoking status was broadly categorized and did not specify smoking exposure duration or intensity. Despite these limitations, the results provide a comprehensive overview of transcriptomic changes, and cancer-associated and smoking-associated biological process differences between lung adenocarcinoma and normal lung tissue.

For future analysis, to further investigate smoking implications on lung adenocarcinoma, creating a dataset in which smoking is categorized by time or intensity of smoking could improve the conclusions of smoking on tumor development. Additionally, incorporating a machine learning model, such as random forest, can help reveal predictive carcinogenic genes to extend beyond DEG analysis.

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GEO Accession Dataset - [GSE40419](#)

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